

# **Analysis of Machining State Patterns from CNC Sensor Data Using Unsupervised Methods**

*Wan Chi Kao MSAI (2027)*  
April 18, 2026

**Department of Khoury College of Computer Sciences  
Northeastern University**

## **ABSTRACT**

**This study proposes an unsupervised learning approach to analyze machining behavior using CNC sensor data. Since labeled data is often unavailable in real manufacturing environments, time series data is divided into fixed length windows and statistical features are extracted. Additional features based on differences between actual and command values are used to describe system deviations. Clustering methods, including K-means, GMM, and SOM, are applied, along with a second level clustering method for experiment level analysis. The results show that SOM achieves the best performance. The method can identify stable, boundary, and unstable machining states, and provides useful information for process monitoring and quality improvement.**

## CONTENTS

|     |                                                     |    |
|-----|-----------------------------------------------------|----|
| I   | Introduction .....                                  | 4  |
| II  | DATA DESCRIPTION .....                              | 7  |
| A   | Dataset Overview .....                              | 7  |
| B   | Raw Data Characteristics .....                      | 7  |
| III | METHODOLOGY .....                                   | 9  |
| A   | Data Preprocessing .....                            | 9  |
| B   | Window Based Representation .....                   | 10 |
| C   | Feature Extraction .....                            | 11 |
| D   | First level Clustering .....                        | 11 |
| E   | Second level Clustering .....                       | 12 |
| F   | Validation and Model Comparison.....                | 13 |
| IV  | Results .....                                       | 14 |
| A   | First Level Clustering Results .....                | 14 |
| B   | Second Level Clustering Results.....                | 16 |
| C   | Window Selection within Each Model .....            | 16 |
| D   | Model Comparison .....                              | 18 |
| V   | Discussion .....                                    | 20 |
| VI  | Conclusions .....                                   | 22 |
|     | References .....                                    | 23 |
|     | Vita .....                                          | 24 |
|     | Appendix .....                                      | 26 |
| A   | Evaluation of Clustering Results Using NMI.....     | 26 |
| B   | Model Selection Strategy Based on Weighted NMI..... | 27 |

## I INTRODUCTION

With the rapid development of smart manufacturing and Industry 4.0, manufacturing is gradually transforming from traditionally depending on experience to data driven decision making, which has become one of the most important ways to improve productivity and product quality. The main point of Industry 4.0 is to construct intelligent factories. By integrating Internet of Things (IoT), big data analysis, automation, and robotics technology, manufacturing processes can automatically perform highly repetitive, high risk, and high precision tasks, which improves overall manufacturing efficiency and quality [1].

In high value industries such as electronics and semiconductors, predictive maintenance, automatic parameter adjustment, and process optimization have already been achieved by using sensors to collect real time production data and combining it with quality monitoring systems and artificial intelligence (AI) techniques. These techniques not only reduce dependence on operators' experience but also push manufacturing systems toward higher levels of automation and intelligence.

The accuracy of mechanical processes is influenced by several factors, including the condition of equipment, the state of process control, and the variation of forces during machining. For example, equipment wears and geometric errors may decrease process accuracy. The selection of machining paths and the setting of feed rates will also affect process control results. On the other hand, variations in machining force and unstable spindle speed may lead to unstable quality. For these reasons, these factors can have a significant impact on final product quality.

However, in traditional processing industries, operations mostly rely on empirical principles and manual decisions, which makes it difficult to maintain stable processing efficiency and product quality. Therefore, it is helpful to improve process stability and system reliability by introducing automation and digital control techniques [2].

In the past, traditional reliability analysis relied on statistical models as the basis for equipment maintenance scheduling and life span management. These models use historical data to predict failure rate ( $\lambda$ ) and mean time to failure (MTTF), and are often combined with the concepts of the Bathtub Curve and Weibull Distribution.

However, traditional approaches still have several limitations in practical applications. Statistical analysis highly depends on data representation and completeness, and insufficient or uneven data may lead to biased results and reduced decision accuracy. In addition, much of the data is derived from expert experience, which may be influenced by subjective factors and reduce objectivity. Furthermore, most traditional approaches are based on static historical data and lack the ability to handle real time conditions, making it difficult to adjust maintenance decisions dynamically. As a result, fixed maintenance strategies

may lead to excessive or insufficient maintenance, which affects machine utilization and overall production efficiency.

Research on the Netherlands manufacturing industry by Erumban (2008) also pointed out that traditional statistical models lack the ability to update and adjust in real time under rapidly changing conditions, which makes it difficult to accurately reflect dynamic production environments [3].

Based on the above analysis, traditional statistical approaches show limitations when applied to reliability evaluation and life span prediction of CNC equipment. Therefore, in recent years, many researchers have introduced AI and machine learning (ML) techniques to improve the accuracy of equipment health prediction and the timeliness of maintenance decisions.

Research by Mr. Tsai (2020) pointed out that processing time is affected by product features, equipment condition, tool wear, and process stability. By combining digital twin technology and ML models to predict tool life span and optimize parameters, it is possible to reduce machine downtime, increase production efficiency, and maintain product quality [4]. When abnormal conditions occur in the machine, they may not only affect process quality but also increase cost and time due to unscheduled machine downtime. Therefore, building a proper equipment maintenance plan is important to maintain precision and ensure stable product quality.

In the machine tool manufacturing process, the condition of tools and dies is an important factor that affects machining quality and production efficiency. Tool wear, dullness, chipping, and die degradation may reduce machining accuracy and increase rework, which further affects process stability and reliability [2]. Previous studies have shown that monitoring tool conditions and replacing tools at the proper time can reduce production costs by about 10% to 40% and improve efficiency by about 10% to 50% [5].

However, in traditional practice, the condition of tools and dies is mainly judged by experienced operators, combined with reliability methods such as MTTF for periodic maintenance. These methods have some limitations. They cannot update data in real time and cannot reflect actual wear under different machining conditions, which leads to limited accuracy and timeliness in tool condition evaluation.

To overcome these limitations, ML has been increasingly applied to tool condition monitoring in recent years. By analyzing sensor data and using learning models, it is possible to improve the accuracy of wear prediction and condition classification. Previous studies have shown that features extracted from machining force, sound, and vibration signals can effectively identify tool wear patterns and improve maintenance strategies [6]. In addition, some models can adjust parameters based on changing machining conditions, which helps improve prediction stability and accuracy [7].

However, in real manufacturing environments, tool conditions change continuously, and it is difficult to obtain real time and accurate labeled data. Instead of directly predicting tool condition, this study focuses on machining process behavior, since these states also reflect system stability and are more difficult to label in real time. Tool wear gradually affects system stability and processing load during machining, which can lead to reduced product quality and reliability.

During CNC machining, process behavior also changes continuously, and different machining states and their occurrence proportions reflect overall process stability. However, these machining states usually lack real time annotations, which makes it difficult to directly apply supervised learning methods.

Therefore, this study focuses on the machining process itself and adopts an unsupervised learning approach. CNC sensor time series data are divided into fixed length windows, where each window represents an instantaneous machining state. These states are grouped using clustering methods, and their distributions are used to describe overall machining stability.

This study aims to explore the underlying patterns of different machining states based on CNC sensor data. To reduce the dependence on labeled data, an unsupervised learning method is used to automatically identify machining states and analyze their relationship with process stability.

The main contribution of this study is to propose a method based on time windows to represent machining states. CNC sensor data are transformed into representative state units, and clustering methods are used to find hidden machining behavior patterns. In addition, this study analyzes the distribution of machining states to build an experiment level stability classification method and further discusses its relationship with tool condition.

The rest of this report is organized as follows. Chapter 2 introduces the data source and preprocessing process. Chapter 3 presents the research method and model design. Chapter 4 shows the experimental results and analysis. Chapter 5 discusses the results and possible improvements. Finally, Chapter 6 presents the conclusion and future work.

## II DATA DESCRIPTION

This section describes the dataset used in this study, including its source, structure, and key characteristics. The data preprocessing steps applied to prepare the dataset for modeling are also presented.

### *A Dataset Overview*

The dataset used in this project comes from a series of CNC milling experiments conducted at the University of Michigan SMART laboratory [8]. A total of 18 machining experiments were performed under different tool conditions, feed rates, and clamping pressures. Each experiment represents a complete machining process on a wax workpiece, including both successful and failed cases. This helps preserve the variability of real machining conditions and improves the representativeness of the dataset.

For each experiment, CNC sensor data were recorded as time series with a sampling rate of 100 milliseconds. The dataset is provided as 18 separate files, where each file corresponds to one experiment. Each file contains sensor measurements from four machine components, including the X axis, Y axis, Z axis, and the spindle. These signals describe the dynamic behavior of the machining process over time.

In addition, the dataset includes experiment level information, such as the tool condition obtained through manual inspection after machining, as well as the feed rate and clamping pressure used in each experiment. In this study, the tool condition is used as reference information for analysis rather than as a direct training label.

### *B Raw Data Characteristics*

The dataset used in this study can be divided into two types: experiment level data and time series sensor data.

The experiment level data record the overall settings and results of each experiment. This includes the experiment ID, material type (wax), feed rate, clamping pressure, and result labels such as tool condition, machining completed and passed visual inspection. These variables are useful for analyzing the relationship between machining stability and product quality.

The time series data are recorded at a sampling interval of 100 milliseconds and describe the operating conditions of machine components during the machining process. The sensor features include both actual and command values of position, velocity, and acceleration, as well as signals such as current, voltage, and power. These data reflect the dynamic load and control behavior of the machine during operation.

In addition, each time step is labeled with a machining process tag. The main machining operations can be divided into six stages, including Layer 1 Up, Layer 1 Down, Layer 2 Up, Layer 2 Down, Layer 3 Up,

and Layer 3 Down. These stages represent the most important and typical parts of the machining process and are the focus of the analysis in this study.

However, the raw time series data also include non-machining periods, such as preparation, tool movement and repositioning, and the end of the process. These segments do not reflect actual machining behavior and show different signal patterns compared to machining periods. As shown in Fig. 1(a), the full time series includes these non-machining segments. In contrast, Fig. 1(b) shows the data after removing them, where the signals are more concentrated and better represent the machining behavior.

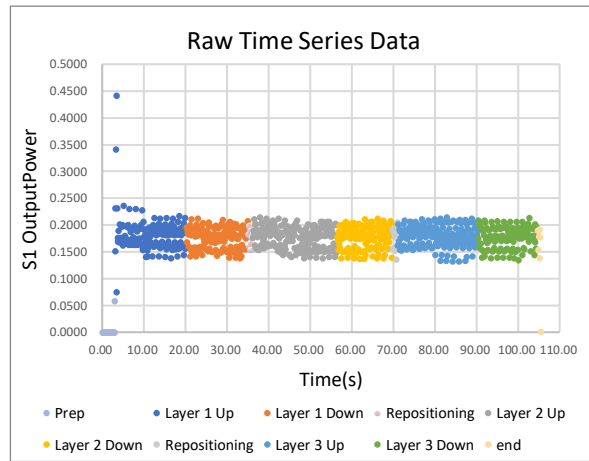


Fig.1(a) Raw Time Series Data

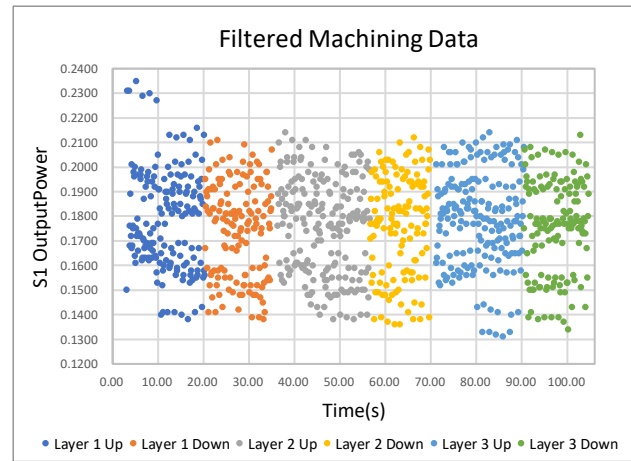


Fig.1(b) Filtered Machining Data

### III METHODOLOGY

This study includes several main steps, including data preprocessing, time series segmentation, feature extraction, clustering model construction, and result validation. First, the raw CNC sensor data are preprocessed to retain only valid machining stages and remove abnormal data.

The data are then divided into fixed length windows, and statistical features are extracted from each window to form feature vectors. Based on these features, three unsupervised learning methods, including K-means, Gaussian Mixture Model (GMM), and Self Organizing Map, are used for clustering analysis.

Finally, second level clustering is performed using K-means with three clusters to represent different machining behavior patterns. Normalized Mutual Information (NMI) is used to evaluate the relationship between the clustering results and actual machining quality.

#### *A Data Preprocessing*

The data preprocessing process in this study includes data loading and validation, filtering of machining stages, removal of abnormal data, feature selection and construction. The goal is to transform the raw CNC time series data into fixed length feature representations for clustering analysis.

First, all experiment files are loaded, and the feature names are checked to ensure consistency across all datasets. This step ensures that all required sensor features are available and avoids issues caused by missing or inconsistent columns.

After checking datasets, abnormal data are removed based on the rules defined in the original dataset documentation. When the data meets specific conditions, such as feed rate equal to 50, X axis actual position equal to 198, and program number not equal to 0, the data point is treated as abnormal and removed. These rules are not defined by this study but come from the original README file. This step helps improve data quality and the reliability of the analysis.

Next, important features are selected from the raw sensor signals. These include actual velocity, actual acceleration, current feedback, output current, output power, and feed rate. In addition, error features are constructed by calculating the difference between actual values and command values. These include position error, velocity error, and acceleration error.

Compared to using only actual values or command values, error features can better reflect whether the machine follows the expected commands. This is important for evaluating machine stability. The selected features used in this study are summarized in Table 1.

| Table 1. Selected features used in this study |                                                    |
|-----------------------------------------------|----------------------------------------------------|
| Category                                      | Features                                           |
| Velocity                                      | X1, Y1, Z1, S1 actual velocity                     |
| Acceleration                                  | X1, Y1, Z1, S1 actual acceleration                 |
| Current                                       | Current feedback and output current                |
| Power                                         | Output power                                       |
| Control                                       | Feed rate                                          |
| Derived                                       | Position error, velocity error, acceleration error |

### B Window Based Representation

Based on the machining process labels, only the main machining stages are kept. These include Layer 1 Up, Layer 1 Down, Layer 2 Up, Layer 2 Down, Layer 3 Up, and Layer 3 Down. While the machine is not machining, such as preparation, repositioning, and the end of the process, are removed. These segments do not reflect actual machining behavior and show different signal patterns. Keeping them may introduce noise and affect the analysis. As shown in Fig. 2, different machining stages exhibit different signal patterns.

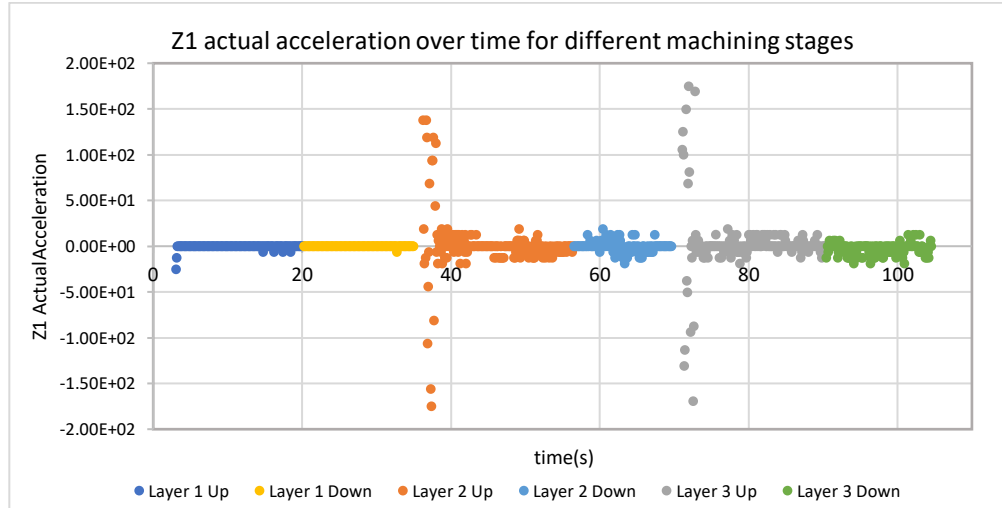


Fig.2 Z1 actual acceleration over time for different machining stages

For window segmentation, the data are divided into fixed length windows. In the initial design, all data were segmented together. However, it was found that different machining stages have different behavior patterns. For example, acceleration signals may show different positive and negative patterns in different stages. Mixing different stages in the same window may reduce the meaning of the features and make clustering more difficult.

Therefore, the window segmentation is performed separately for each machining process. This avoids mixing data from different stages and improves the consistency of the features.

To compare different time scales, three window sizes are used, including 10, 20, and 30. A 50 percent overlap is applied between consecutive windows to keep continuity. If a process segment is shorter than one full window but longer than half of the window size, it is still kept as a valid window. This design helps preserve useful information from shorter segments while avoiding excessive fragmentation.

### *C Feature Extraction*

After segmentation, the raw time point values are not used directly. Instead, each time window is transformed into a fixed length feature vector so that different machining states can be compared in a consistent way.

For each window, statistical features are calculated to summarize the signal behavior within a short period. These include mean, mean absolute difference, standard deviation, maximum, minimum, range, and root mean square. While basic statistics such as mean, maximum, and minimum describe the general level of the signals, they may not fully capture the dynamic characteristics of the machining process.

To better represent the variation and energy of the signals, additional features such as range and root mean square are included. The range reflects the amplitude of change within a window, while the root mean square captures the overall signal energy. These features help describe the dynamic behavior of the machining process more effectively.

By converting each window into a fixed dimensional representation, all windows can be compared under the same structure. This makes the data suitable for clustering analysis and helps capture the dynamic characteristics of the machining process.

Finally, all features are standardized to ensure similar scales, which improves the stability and reliability of the clustering results.

### *D First level Clustering*

In this study, the time series data are divided into fixed length windows. Each window is then transformed into a feature vector. To avoid mixing different machining behaviors, the data are separated by machining process, and clustering is performed for each process independently. This helps the model learn more consistent patterns within each machining stage.

Three unsupervised models are used in this study, including K-means, Gaussian Mixture Model, and Self Organizing Map. All models use the same features and the same preprocessing steps to ensure a fair comparison.

K-means is a clustering method that groups data points based on the distance between them. It assigns each data point to the nearest cluster center, and each point belongs to only one cluster. In this study,

window level feature vectors are used as input. Metadata columns such as the id of experiment, the id of window, and window size, and Machining Process are removed before training. All features are standardized, and Principal Component Analysis is optionally applied to keep 90 percent of the variance. To choose the number of clusters, values from 2 to 9 are tested. The Silhouette Score is used as the main evaluation method because it measures how well the clusters are separated and how close the data points are within each cluster. Davies Bouldin Index and Calinski Harabasz Index are also recorded as additional references. The final number of clusters is selected based on the best overall performance.

Gaussian Mixture Model is a clustering method that models the data using several Gaussian distributions. Each data point has a probability of belonging to each cluster, which allows the model to describe overlapping clusters. The same input features and preprocessing steps are used as in K-means. Metadata columns are removed, and all features are standardized. Principal Component Analysis is also applied when needed. To determine the number of clusters, values from 2 to 9 are tested. Unlike K-means, this model focuses on how well the data fits a statistical distribution. Therefore, Bayesian Information Criterion is used as the main method for model selection. This criterion considers both how well the model fits the data and how complex the model is. It helps prevent overfitting when the number of clusters becomes too large. Akaike Information Criterion, Silhouette Score, Davies Bouldin Index, and Calinski Harabasz Index are also recorded for additional comparison.

Self-Organizing Map is a neural network that maps high dimensional data into a low dimensional grid. It keeps similar data points close to each other, which makes it easier to observe relationships between different machining states. The same feature set and preprocessing steps are used as in the other models. Instead of choosing the number of clusters directly, this method requires defining a grid structure. In this study, several grid sizes are tested, including  $2 \times 1$ ,  $3 \times 1$ ,  $4 \times 1$ ,  $2 \times 2$ . After training, each data point is assigned to its best matching unit on the grid, and this position is used as the cluster label. To select the best grid structure, the Silhouette Score is used as the main evaluation method. Davies Bouldin Index and Calinski Harabasz Index are also used for comparison. The grid with the best performance is selected for each machining process.

### *E Second level Clustering*

After the first level clustering, the results are transformed into experiment level features to analyze overall machining stability. For each window, the cluster label, the id of experiment, and Machining Process are kept. Then, for each experiment and each machining process, the proportion of each cluster is calculated. These proportions are used to describe the machining behavior at different stages.

Next, the cluster ratios from all machining processes are combined into a fixed length feature vector for each experiment. This representation keeps the information from different stages and reflects the overall behavior of the experiment.

After building the experiment level features, a second level clustering is performed. In this study, K-means is used, and the number of clusters is fixed at 3. This is based on the research goal, which is to classify the experiments into three types of machining stability, including stable, boundary, and unstable conditions. The purpose of this step is not to test different numbers of clusters, but to provide a clear and meaningful classification of overall machining behavior.

Before clustering, all features are standardized to make sure that different scales do not affect the results. The final clustering results are used as behavior labels for each experiment and are used for further analysis.

### *F Validation and Model Comparison*

The second level clustering results are compared with actual labels to understand the meaning of each behavior cluster. For machining finalized, passed visual inspection, and tool condition, the proportion of each label category within each cluster is examined. This allows the interpretation of each cluster based on machining completion, product quality, and tool condition.

In addition, Normalized Mutual Information is used to measure the overall consistency between clustering results and the actual labels. For each window size, NMI is calculated for machining finalized, passed visual inspection, and tool condition.

The model selection process is designed as a two step procedure. In the first step, for each clustering model, different window sizes are compared. To combine the evaluation results, a weighted score is calculated based on the three NMI values. In this study, the weights are set to 0.6 for machining finalized, 0.2 for passed visual inspection, and 0.2 for tool condition. The best window size is then selected based on the highest weighted score.

In the second step, the best result from each model is collected and compared using the same weighted score. The results from K-means, Gaussian Mixture Model, and Self Organizing Map are ranked, and the model with the highest score is selected as the overall best clustering model.

## IV RESULTS

This section presents the results of the proposed two stage clustering framework. The analysis includes the first level clustering of time window data, followed by experiment level clustering based on distribution patterns. In addition, the optimal window size is determined for each model, and a comparison among clustering models is conducted.

### A First Level Clustering Results

The first level clustering results are shown in Table 2 to Table 4. For each model, the optimal clustering structure is determined for each machining process under different window sizes.

For K-means, the optimal number of clusters varies across machining processes and window sizes. In most cases, smaller values of  $k$  are selected for early layers, while larger values are observed in later layers, especially for Layer 3. The clustering results also change with different window sizes, indicating that the temporal resolution affects the grouping structure.

| Table 2. Optimal Cluster Number for K-means |                   |        |                  |         |
|---------------------------------------------|-------------------|--------|------------------|---------|
| Window Size                                 | Machining Process | Best K | Silhouette Score | Windows |
| 10                                          | Layer 1 Down      | 2      | 0.442            | 461     |
|                                             | Layer 1 Up        | 2      | 0.424            | 675     |
|                                             | Layer 2 Down      | 2      | 0.518            | 448     |
|                                             | Layer 2 Up        | 3      | 0.454            | 567     |
|                                             | Layer 3 Down      | 5      | 0.355            | 424     |
|                                             | Layer 3 Up        | 6      | 0.349            | 497     |
| 20                                          | Layer 1 Down      | 2      | 0.427            | 218     |
|                                             | Layer 1 Up        | 2      | 0.376            | 324     |
|                                             | Layer 2 Down      | 2      | 0.466            | 213     |
|                                             | Layer 2 Up        | 3      | 0.434            | 273     |
|                                             | Layer 3 Down      | 2      | 0.471            | 201     |
|                                             | Layer 3 Up        | 8      | 0.307            | 239     |
| 30                                          | Layer 1 Down      | 4      | 0.202            | 138     |
|                                             | Layer 1 Up        | 2      | 0.363            | 208     |
|                                             | Layer 2 Down      | 4      | 0.398            | 135     |
|                                             | Layer 2 Up        | 5      | 0.324            | 175     |
|                                             | Layer 3 Down      | 6      | 0.313            | 127     |
|                                             | Layer 3 Up        | 2      | 0.430            | 150     |

For Gaussian Mixture Model, the selected number of clusters is generally higher compared to K-means. This is because GMM allows more flexible cluster shapes and is selected based on BIC. The results show that more clusters are used to capture complex patterns in the data. The detailed results are summarized in Table 3.

For Self-Organizing Map, the selected grid structure is relatively simple. Most cases use a 2 by 1 grid,

resulting in a small number of neurons. This suggests that the data are mapped into a low dimensional representation, which focuses on preserving the overall structure rather than increasing cluster complexity. The corresponding results are summarized in Table 4.

Overall, the clustering structure varies across models, window sizes, and machining processes, reflecting different ways of representing machining behavior at the first level.

| Table 3. Optimal Cluster Number for GMM |                   |        |          |         |
|-----------------------------------------|-------------------|--------|----------|---------|
| Window Size                             | Machining Process | Best K | BIC      | Windows |
| 10                                      | Layer 1 Down      | 9      | 39012.08 | 461     |
|                                         | Layer 1 Up        | 7      | 63355.92 | 675     |
|                                         | Layer 2 Down      | 5      | 35962.6  | 448     |
|                                         | Layer 2 Up        | 7      | 46176.21 | 567     |
|                                         | Layer 3 Down      | 7      | 33441.3  | 424     |
|                                         | Layer 3 Up        | 9      | 38433.75 | 497     |
| 20                                      | Layer 1 Down      | 9      | 15992.5  | 218     |
|                                         | Layer 1 Up        | 9      | 30596.7  | 324     |
|                                         | Layer 2 Down      | 9      | 14954.1  | 213     |
|                                         | Layer 2 Up        | 9      | 21778.59 | 273     |
|                                         | Layer 3 Down      | 9      | 15718.11 | 201     |
|                                         | Layer 3 Up        | 6      | 18155.2  | 239     |
| 30                                      | Layer 1 Down      | 6      | 10425    | 138     |
|                                         | Layer 1 Up        | 9      | 19244.16 | 208     |
|                                         | Layer 2 Down      | 8      | 9663.478 | 135     |
|                                         | Layer 2 Up        | 8      | 12715.07 | 175     |
|                                         | Layer 3 Down      | 8      | 7072.48  | 127     |
|                                         | Layer 3 Up        | 8      | 11683.87 | 150     |

| Table 4. Optimal SOM Grid Structure |                   |        |        |         |         |
|-------------------------------------|-------------------|--------|--------|---------|---------|
| Window Size                         | Machining Process | Grid X | Grid Y | Neurons | Windows |
| 10                                  | Layer 1 Down      | 2      | 1      | 2       | 461     |
|                                     | Layer 1 Up        | 2      | 1      | 2       | 675     |
|                                     | Layer 2 Down      | 2      | 1      | 2       | 448     |
|                                     | Layer 2 Up        | 3      | 1      | 3       | 567     |
|                                     | Layer 3 Down      | 2      | 1      | 2       | 424     |
|                                     | Layer 3 Up        | 2      | 1      | 2       | 497     |
| 20                                  | Layer 1 Down      | 2      | 1      | 2       | 218     |
|                                     | Layer 1 Up        | 2      | 1      | 2       | 324     |
|                                     | Layer 2 Down      | 2      | 1      | 2       | 213     |
|                                     | Layer 2 Up        | 2      | 1      | 2       | 273     |
|                                     | Layer 3 Down      | 2      | 1      | 2       | 201     |
|                                     | Layer 3 Up        | 2      | 1      | 2       | 239     |
| 30                                  | Layer 1 Down      | 2      | 1      | 2       | 138     |
|                                     | Layer 1 Up        | 2      | 1      | 2       | 208     |
|                                     | Layer 2 Down      | 2      | 1      | 2       | 135     |
|                                     | Layer 2 Up        | 2      | 1      | 2       | 175     |
|                                     | Layer 3 Down      | 2      | 1      | 2       | 127     |
|                                     | Layer 3 Up        | 2      | 1      | 2       | 150     |

## B Second Level Clustering Results

Table 5 presents the distribution of experiments in the second level clustering for all models when the window size is 30. In total, 18 experiments are grouped into three behavior clusters, but the distribution across clusters differs among the models.

| Table 5. Distribution of Experiments in Second Level Clusters (Window Size = 30) |         |     |     |
|----------------------------------------------------------------------------------|---------|-----|-----|
| Cluster/model                                                                    | K-means | GMM | SOM |
| 0                                                                                | 7       | 3   | 4   |
| 1                                                                                | 10      | 14  | 10  |
| 2                                                                                | 1       | 1   | 4   |

For K-means, most experiments are assigned to Cluster 1, which contains 10 experiments, while Cluster 0 and Cluster 2 contain 7 and 1 experiments, respectively. This indicates that the clustering result is relatively concentrated in one dominant cluster.

For Gaussian Mixture Model, a similar pattern is observed. A large number of experiments, 14 in total, are grouped into Cluster 1, while Cluster 0 and Cluster 2 contain only 3 and 1 experiments. Compared to K-means, the distribution is even more concentrated.

In contrast, Self-Organizing Map shows a more balanced distribution. The numbers of experiments in the three clusters are 4, 10, and 4, respectively. This suggests that SOM produces a more evenly distributed clustering structure at the experiment level.

Overall, the results show that different first level clustering methods lead to different distributions in the second level clustering. K-means and GMM tend to group most experiments into a single dominant cluster, while SOM provides a more balanced grouping. These differences will be further evaluated in the following sections.

## C Window Selection within Each Model

This section compares the performance of each clustering model under different window sizes and determines the optimal configuration for each model.

To ensure a consistent comparison, a weighted score is used to summarize the evaluation results based on three NMI metrics, including machining finalized, passed visual inspection, and tool condition. The weights are assigned as 0.6, 0.2, and 0.2, respectively, to reflect their relative importance.

For K-means, the results show that window sizes 10 and 20 achieve identical performance, with the same weighted score of 0.39498, as shown in Table 6. Although window size 30 yields a higher NMI value for inspection, its overall score is slightly lower. In addition, the distribution patterns observed in Fig. 3

remain similar across different window sizes, indicating that the impact of temporal resolution on K-means is limited. Therefore, window size 10 is selected as the optimal configuration.

| Table 6. NMI Scores of K-means under Different Window Sizes |                 |                  |                      |
|-------------------------------------------------------------|-----------------|------------------|----------------------|
| Window size                                                 | NMI (Finalized) | NMI (Inspection) | NMI (Tool Condition) |
| <b>10</b>                                                   | <b>0.467842</b> | <b>0.359948</b>  | <b>0.211435</b>      |
| 20                                                          | 0.467842        | 0.359948         | 0.211435             |
| 30                                                          | 0.381659        | 0.539443         | 0.260504             |



Fig.3 Comparison of Behavior Clusters and Inspection Results across Window Sizes (K-means)

For GMM, window size 30 achieves the highest weighted score of 0.18182, as shown in Table 7. Compared to other configurations, it provides a more balanced performance across different evaluation metrics. The patterns observed in Fig. 4 also show noticeable differences across window sizes, suggesting that GMM is more sensitive to temporal resolution. Thus, window size 30 is selected.

For SOM, window size 30 significantly outperforms the other configurations, achieving the highest weighted score of 0.49403, as shown in Table 8. As shown in Fig. 5, the relationship between behavior clusters and inspection results becomes clearer under this configuration, indicating that a larger window size better captures the overall machining behavior. Therefore, window size 30 is selected for SOM.

Overall, the optimal window sizes are determined as follows: K-means uses window size 10, GMM uses window size 30, and SOM uses window size 30. These configurations are used for subsequent model comparison.

| Table 7. NMI Scores of GMM under Different Window Sizes |                    |                    |                      |
|---------------------------------------------------------|--------------------|--------------------|----------------------|
| Window size                                             | NMI (Finalized)    | NMI (Inspection)   | NMI (Tool Condition) |
| 10                                                      | 0.173696694        | 0.202050634        | 0.063519592          |
| 20                                                      | 0.210868868        | 0.058711693        | 0.13660743           |
| <b>30</b>                                               | <b>0.176702612</b> | <b>0.209982496</b> | <b>0.169024509</b>   |



Fig.4 Comparison of Behavior Clusters and Inspection Results across Window Sizes (GMM)

| Table 8. NMI Scores of SOM under Different Window Sizes |                    |                    |                      |
|---------------------------------------------------------|--------------------|--------------------|----------------------|
| Window size                                             | NMI (Finalized)    | NMI (Inspection)   | NMI (Tool Condition) |
| 10                                                      | 0.723874715        | 0.159834878        | 0.008561932          |
| 20                                                      | 0.54620636         | 0.202050634        | 0.063519592          |
| <b>30</b>                                               | <b>0.694818218</b> | <b>0.343323954</b> | <b>0.042360699</b>   |

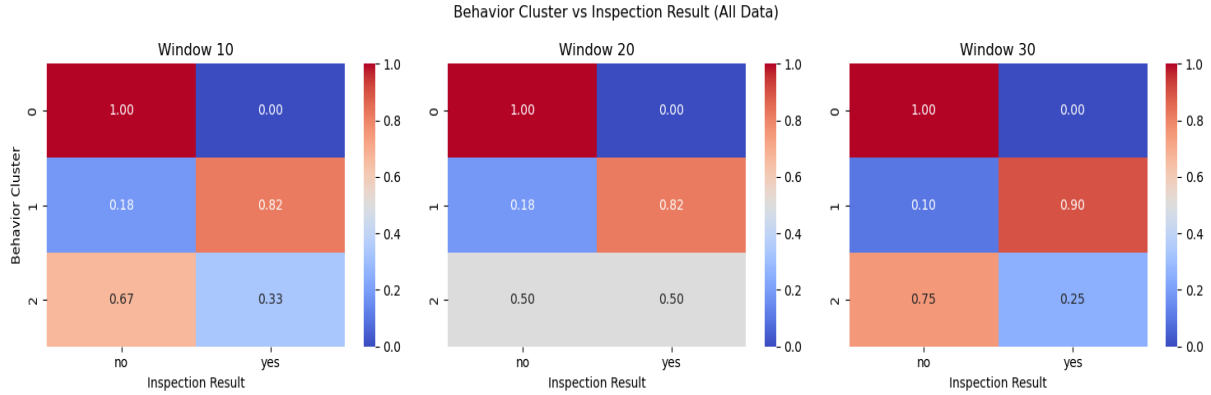


Fig.5 Comparison of Behavior Clusters and Inspection Results across Window Sizes (SOM)

## D Model Comparison

The performance of different clustering models is compared based on their optimal window configurations. Table 9 summarizes the best results for each model under their selected window sizes.

Among all models, SOM achieves the highest final score of 0.494, indicating the best overall performance. K-means shows moderate performance, with a final score of 0.395, while GMM has the lowest score of 0.182.

The superior performance of SOM suggests that it is more effective in capturing the underlying machining behavior patterns. In particular, SOM achieves a significantly higher NMI value for machining

finalized, indicating a stronger ability to distinguish overall machining outcomes. Therefore, SOM with window size 30 is selected as the final model for this study.

| Model   | Window Size | NMI (Finalized) | NMI (Inspection) | NMI (Tool Condition) | Final Score |
|---------|-------------|-----------------|------------------|----------------------|-------------|
| K-means | 10          | 0.4678          | 0.3599           | 0.2114               | 0.3950      |
| GMM     | 30          | 0.1767          | 0.2100           | 0.1690               | 0.1818      |
| SOM     | 30          | 0.6948          | 0.3433           | 0.0424               | 0.4940      |

In addition to model comparison, the clustering results of the selected SOM model are further analyzed to understand the meaning of each cluster. Fig.6(a) and Fig.6(b) show the relationship between clusters and machining outcomes.

From the machining completed results, Cluster 0 is associated with unfinished machining, while Cluster 1 and Cluster 2 both correspond to completed machining. This indicates that Cluster 0 represents unstable behavior where the machine cannot complete the process.

However, further analysis using inspection results shows a clear difference between Cluster 1 and Cluster 2. Cluster 1 has a high proportion of passed inspection results, which indicates stable machining performance. In contrast, Cluster 2 has a high proportion of failed inspection results, even though the machining process is completed.

Based on these observations, Cluster 1 can be considered as stable behavior, Cluster 0 represents unstable behavior, and Cluster 2 represents boundary behavior. This boundary cluster reflects cases where the machine can complete the process but cannot guarantee product quality.

This result demonstrates that the clustering method can distinguish not only whether machining is completed, but also the quality of the outcome. It provides a more detailed understanding of machining behavior, which is important for practical applications.



Fig.6(a) Cluster vs Machining Completion

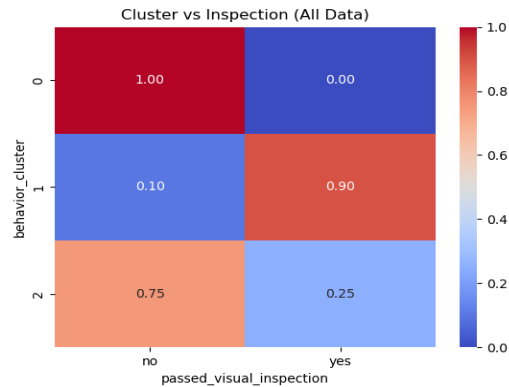


Fig.6(b) Cluster vs Inspection Results

## V DISCUSSION

This study discusses the clustering results and explains the differences between methods. The main goal is to support smart manufacturing. In traditional machines, it is hard to get clear labels from data. For this reason, unsupervised learning is used to find patterns from sensor signals.

First, data preprocessing is very important. The original dataset includes both machining data and data when the machine is not machining. If these data are mixed together, the clustering results will become unclear. In this study, the data is separated based on Machining Process. Data that is not related to actual machining is removed. This step reduces noise and keeps only useful information. The results become more stable after this step.

Second, this study uses a sliding window method. This method converts time series data into multiple samples and keeps time information. The window size has a strong effect on the results. A small window makes the features unstable. A large window may hide important behavior in a short time. Several window sizes are tested in this study. The results show that different models work best with different window sizes. This means each method uses time information in a different way.

Third, feature design is very important. Raw sensor data is not enough to clearly separate machining behavior. To improve this, new features are created. These features are based on the difference between actual values and command values. They include position, velocity, and acceleration differences. These values reflect control errors in the machine and system deviation. They are closely related to machining stability. This idea was not used at the beginning. It was added later after observing that the clustering results were not clear. After adding these features, the clusters become more distinct. This shows that feature design has a strong impact on the results.

In model comparison, SOM shows the best performance. K-means is second, and GMM has the lowest score. This suggests that the data has a complex structure. SOM can keep the structure of the data and capture complex patterns. K-means works better when the data is simple. GMM needs to estimate data distribution. In this study, the data is limited and complex, so GMM does not perform well.

Another challenge is how to understand the clustering results. This study uses unsupervised learning, so there are no true labels. It is not easy to know what each cluster means. To solve this problem, the study looks at the distribution and statistics of each cluster. It also compares the results with machining completion and inspection results at the experiment level. This helps explain what each cluster represents and how it relates to machining stability.

Based on the analysis of clustering results, three types of machining behavior can be identified. Cluster 0 is associated with unfinished machining, which represents unstable behavior. Cluster 1 shows a high

proportion of completed machining and passed inspection results, indicating stable behavior. Cluster 2 also corresponds to completed machining, but many cases fail inspection. This indicates that Cluster 2 represents boundary behavior, where the process can be completed but product quality is not stable.

This result shows that the clustering method can provide more detailed information than a simple success or failure classification. It helps distinguish between different levels of machining performance, which is useful for identifying potential quality risks in real applications.

In conclusion, this study shows that data preprocessing, feature design, and time representation are very important. Even without labeled data, useful patterns can still be found from sensor signals. This approach can support smart manufacturing. In the future, more data and more sensors can be used. Deep learning methods can also be tested to improve performance.

## VI CONCLUSIONS

This study addresses the problem that traditional manufacturing often relies on experience and lacks labeled data. An unsupervised learning approach is proposed to analyze machining behavior and evaluate process stability using CNC sensor data. In real manufacturing environments, machining behavior changes over time, and labeled data is often not available. Because of this, supervised learning methods are difficult to apply. Therefore, this study focuses on machining behavior and uses a data driven approach to identify patterns. This study also proposes a systematic framework to evaluate clustering results and select optimal configurations based on multiple quality related criteria.

In this study, CNC time series data is divided into fixed length windows. Each window represents a period of machining behavior. Statistical features are extracted from each window. In addition, new features are created based on the difference between actual values and command values. These features describe system errors and deviations during machining. After feature extraction, three clustering methods, K-means, GMM, and SOM, are applied. A second level clustering approach is also used to connect window level results with experiment level stability. This design allows the model to capture both local behavior patterns and overall process characteristics.

The results show that data preprocessing plays an important role in clustering performance. After removing data outside the actual machining process, noise is reduced and the results become more stable and easier to understand. The choice of window size also has a strong effect on the results. A small window makes the features unstable, while a large window may hide important changes within the process. Different window sizes capture machining behavior at different temporal resolutions, which directly affects pattern separability. In model comparison, SOM achieves the best overall performance, which indicates that it can better capture complex machining behavior.

Further analysis of the clustering results shows that machining behavior can be divided into three types: stable, boundary, and unstable. The stable type is associated with completed machining and passed inspection. The unstable type is related to unfinished machining. The boundary type can complete the machining process, but often fails inspection. This means that the process is not reliable in terms of quality. This finding shows that the proposed method can provide more detailed information than a simple success or failure classification.

This study contributes a practical unsupervised learning framework that links machining behavior patterns with quality outcomes, enabling both evaluation and interpretation of clustering results. In conclusion, this study demonstrates that meaningful machining patterns can be identified from sensor data without labeled information. The proposed method can be used to analyze machining stability and has

potential for smart manufacturing applications. It can also support early detection of quality risks and process improvement. In future work, more data and more sensor information can be included to improve model performance. Deep learning methods and real time monitoring systems can also be explored for further development.

## REFERENCES

- [1] M. Rüßmann et al., “Industry 4.0: The Future of Productivity and Growth in Manufacturing Industries,” Boston Consulting Group, 2015.
- [2] J. Yu, Z. Lu, and Y. Liu, “簡析機械加工對加工精度的影響 [Analysis of the Influence of Machining on Machining Accuracy],” *Mechanical and Electronic Control Engineering*, vol. 4, no. 11, pp. 64–66, Jan. 2022.
- [3] A. A. Erumban, “Lifetimes of Machinery and Equipment: Evidence from Dutch 106 Manufacturing,” *Review of Income and Wealth*, vol. 54, no. 2, pp. 237–268, Jun. 2008.
- [4] Y.-T. Tsai, “等負載加工之參數訂定策略 [Parameter Determination Strategy for Equal-Load Machining],” M.S. thesis, Dept. of Mechanical Engineering, National Taiwan University, Taipei, Taiwan, Jul. 2020.
- [5] P. Li, Y.-X. Huang, P.-C. Xia, and L. Shi, “基於一維卷積長短期記憶網路的多訊號融合刀具磨損評估 [Multi-Signal Fusion Tool Wear Evaluation Based on One-Dimensional Convolutional LSTM Network],” *Mechanical and Electronic Engineering*, vol. 39, no. 5, pp. 8–14, 2021.
- [6] Q. Wang, H. Wang, L. Hou, and S. Yi, “Overview of Tool Wear Monitoring Methods Based on Convolutional Neural Network,” *Applied Sciences*, vol. 11, no. 24, p. 12041, Jan. 2021.
- [7] Z. Ma, M. Zhao, X. Dai, and Y. Chen, “A Hybrid-driven Probabilistic State Space Model for Tool Wear Monitoring,” *Mechanical Systems and Signal Processing*, vol. 200, pp. 110599–110599, Jul. 2023.
- [8] CNC Milling Dataset – University of Michigan SMART Lab, Kaggle  
<https://www.kaggle.com/datasets/shasun/tool-wear-detection-in-cnc-mill>

## VITA

Wan Chi Kao is currently pursuing a Master's degree in Artificial Intelligence at Northeastern University. She received her Master's degree in Mechanical Engineering from National Taiwan University, where she conducted research in a reliability laboratory. Her thesis applied machine learning to predict press brake pressure and improve process stability. She received her Bachelor's degrees in Civil Engineering and Mechanical Engineering from National Yang Ming Chiao Tung University. Her research interests include machine learning, smart manufacturing, and reliability engineering, focusing on applying data driven methods to improve manufacturing processes.

## APPENDIX

### *A Evaluation of Clustering Results Using NMI*

This function computes the Normalized Mutual Information (NMI) between clustering results and ground truth labels, including machining completion, inspection outcome, and tool condition.

```
# Using NMI to compare the consistency of the clustering results with the actual quality
def compute_nmi_scores(experiment_result, window_size):
    # Load ground truth labels
    train_df = load_data("train.csv")
    train_df["experiment_id"] = train_df["No"].apply(lambda x: f"experiment_{int(x):02d}")

    label_df = train_df[
        ["experiment_id", "machining_finalized",
         "passed_visual_inspection", "tool_condition"]
    ]

    # Merge clustering results with labels
    merged = experiment_result.merge(label_df, on="experiment_id", how="left")

    # NMI for machining completion
    nmi_finalized = normalized_mutual_info_score(
        merged["machining_finalized"],
        merged["behavior_cluster"]
    )

    # NMI for inspection (finished only)
    finished = merged[merged["machining_finalized"] == "yes"]
    nmi_inspection = normalized_mutual_info_score(
        finished["passed_visual_inspection"],
        finished["behavior_cluster"]
    )

    # NMI for tool condition
    nmi_tool = normalized_mutual_info_score(
        finished["tool_condition"],
        finished["behavior_cluster"]
    )
```

```

return pd.DataFrame({
    "window_size": [window_size],
    "nmi_finalized": [nmi_finalized],
    "nmi_inspection_finished_only": [nmi_inspection],
    "nmi_tool_condition": [nmi_tool]
})

```

Listing A1 Evaluation of Clustering Results Using NMI

### *B Model Selection Strategy Based on Weighted NMI*

This function selects the optimal window size based on weighted NMI metrics.

```

## compare three different window sizes and find the one that can best capture the differences
in machining behavior patterns across different experiments and different machining processes.
def select_best_window(nmi_df, model_name):
    result = nmi_df.copy()

    result["final_score"] = (
        0.6 * result["nmi_finalized"] +
        0.2 * result["nmi_inspection_finished_only"] +
        0.2 * result["nmi_tool_condition"]
    )

    result["model_name"] = model_name

    best_row = result.sort_values(
        by="final_score",
        ascending=False
    ).reset_index(drop=True).iloc[0]

    return result, best_row

```

Listing B1 Model Selection Strategy Based on Weighted NMI